

C. Annoying Game

time limit per test: 2 seconds
 memory limit per test: 256 megabytes

You are given two integer arrays a and b , both of length n , and a total number of turns k .

Alice and Bob play a game by taking turns modifying array a . Alice goes first. The game lasts for exactly k turns.

On their turn, a player must choose an index i ($1 \leq i \leq n$) and perform **one** of the following operations:

- **Add:** Increase a_i by b_i , i.e., set $a_i := a_i + b_i$.
- **Subtract:** Decrease a_i by b_i , i.e., set $a_i := a_i - b_i$.

After the k -th turn is complete, the final score is calculated as the *maximum non-empty subarray sum* of the modified array a . Alice's goal is to maximize the final score, while Bob's goal is to minimize the final score.

Assuming both players play optimally to achieve their respective goals, determine the final score.

The *maximum non-empty subarray sum* of a is defined as $\max_{1 \leq i \leq j \leq n} S(i, j)$, where $S(i, j) = a_i + a_{i+1} + \dots + a_j$. Note that empty subarrays are not considered.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and k ($1 \leq n \leq 2 \cdot 10^5$, $1 \leq k \leq 2 \cdot 10^5$) — the length of the arrays and the total number of turns.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($-10^9 \leq a_i \leq 10^9$) — the elements of array a .

The third line contains n integers $b_1, b_2, \dots, b_n$ ($0 \leq b_i \leq 10^9$) — the elements of array b .

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print a single integer — the final score after k turns, assuming both players play optimally.

Example

input

Copy

```
5
5 200000
3 -1 9 -5 4
0 0 0 0 0
4 5
10 10 10 10
1 1 1 1
3 1
2 -7 3
1 11 3
3 2
2 -7 3
1 11 3
1 1
-3
```

Codeforces Round 1067 (Div. 2)

Contest is running

01:11:08

Contestant



→ Submit?

Language: GNU G++17 7.3.0

Choose file: No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

→ Score table

	Score
Problem A	404
Problem B	1010
Problem C	1212
Problem D	1616
Problem E	2222
Problem F1	2020
Problem F2	808
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:48 from the first attempt

2

output

Copy

11
41
9
3
-1

Note

For the first test case:

- Any move can't change the array a because all elements in b are zero.
- The maximum non-empty subarray sum of this final array is $3 + (-1) + 9 = 11$.

For the second test case, one of the possible optimal sequences of moves is:

- Alice adds $b_3 = 1$ to a_3 . Array a becomes $[10, 10, 11, 10]$.
- Bob subtracts $b_1 = 1$ from a_1 . Array a becomes $[9, 10, 11, 10]$.
- Alice adds $b_3 = 1$ to a_3 . Array a becomes $[9, 10, 12, 10]$.
- Bob subtracts $b_1 = 1$ from a_1 . Array a becomes $[8, 10, 12, 10]$.
- Alice adds $b_4 = 1$ to a_4 . Array a becomes $[8, 10, 12, 11]$.
- The maximum non-empty subarray sum of this final array is $8 + 10 + 12 + 11 = 41$.

For the third test case, one of the possible optimal sequences of moves is:

- Alice adds $b_2 = 11$ to a_2 . Array a becomes $[2, 4, 3]$.
- The maximum non-empty subarray sum of this final array is $2 + 4 + 3 = 9$.

For the fourth test case, one of the possible optimal sequences of moves is:

- Alice adds $b_2 = 11$ to a_2 . Array a becomes $[2, 4, 3]$.
- Bob subtracts $b_2 = 11$ from a_2 . Array a becomes $[2, -7, 3]$.
- The maximum non-empty subarray sum of this final array is 3.

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